Exp 1 Study of various Network command Used In linux and windows

1 arp-a Show IP address of your computer user along with IP address and MAC address of your router.

O/P

```
Interface 192.168.156.1 ... 0x7
 Internet address  Physical address  Type
  192.168.56.255   ff-ff-ff-ff-ff    Static
  244.0.0 22 .     01-00-5e-002      Static
  23.9.255.25.250  01-00-5e-ff-ff    Static
```

2 hostname: Display name of your computer

O/P: blackie

3 ipconfig/all: Display detailed configuration information about TCP/IP connection and type of ethernet adapter in the System

O/P

Window IP configuration

Host name . . . . blackie

Primary DNs Suffix

Node type

4 nbstat -a : Display protocal Statistic and curr
TCP/IP connection using NBT

O/P

```
NBTSTAT [[-a Remote Name] [- A IP address ] [
```

5 netstat Display a variety of Statistics abou
a computer active TCP/IP connection

O/P

```
Active connection

Proto  Local address         Foreign address          State
TCP    10.76.96.60:49175     10:76.96.163:domain      Timewai
TCP    10.76.96.60 :52239    142.250.183.http         Establishe
TCP    10.76.96.60 :56954    10.76.96.113:domain      Time-
```

6 nslookup: Display DNs detail such as IP Address of the pariticalar compertor

O/p :-

Server : Unkown
Address : 10.76.196.163

2 Path ping : Traces the root to destinction address the lanches a 25 second test of each routes along the way

O/P

```
Pathping [-g host-list] [-h maximum_hops]
[-i address]
```

Linux Networking Commands

1 ip : Help form setting up new system and assiging IP to trouble shooting escisting system.

2 If config : The ifconfig is staple in many Sys admin's tool helt for configiuring trouble shooting networks.

3 mtr: MY TR is program with CLI that are a network diagnostic and troubleshoting tool.

→ tcp.dump: The tcpdump command is designed for capturing and displaying packets.

→ Ping: Ping is tool that verifies IP-level connectivity to another TCP/IP computer by sending Internet control Messag Protocol (ICMP) Echo Request Message Protocol (ICMP) Echo Request Message

6 If config

```
lo: flags: 73 <UP: LOOK back, RUNNING> mtu6553
  inet 127.0.0.1 netmask 255.0.0.0
  inet6 ::1 prefix len 125 scope idx<localhost>
```

6 mtr → o/p

```
        My traceover [v0.87]
  local host. local domin (::)
Key   Help   Display mode  Restat   statistic
                           Packets
Host                        loss%     last
1::1                        0.0%      0.2
```

4 tcpdump

```
listening on enp7s0f4u1

IP6 93.243.107.34.bc.googleusercontent.com
.http > blackie.56518, seq 5929, ack 35457, window
```

5 Ping

```
Ping 8.8.8.8
64 bytes from 8.8.8.8 icmp seq=1 ttl=113
time = 37.4 ms
```

Result:
Thus, the various command network used in linux and window are executed successfully